

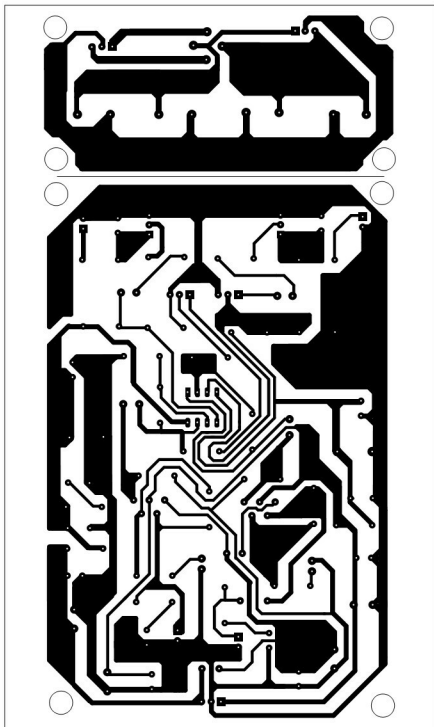


Fig. 4: A single-side, actual-size PCB layout for stereo brick amplifier

Construction

Construction steps involve PCB design, mounting of components on the PCB, cabinet design, connecting the power supply to PCB and extending wires from the PCB to the front and rear panels.

Having conventional circuitry, the amplifier has to be squeezed into a low-cost brick shaped box. 21V-0-21V/2A power transformer occupies almost half of the internal space. To make room for the PCB of amplifier circuit in the brick cabinet, part of

power supply section is mounted on a separate PCB so that it can be fixed on the rear side of the cabinet. Pin details of IC LM1875 are shown in Fig. 3. Note that proper finned heatsinks should be used for power amplifier ICs (LM1875). While designing PCB layout, there is need to determine a proper location on the board to mount the heatsinks for IC2 and IC3.

A single-side, actual-size PCB is shown in Fig. 4 and its component layout in Fig. 5. The PCB for the power supply section can be separated from the amplifier section by cutting along the straight line shown in the Fig. 4 and 5 layouts.

The construction steps should be followed in the following sequence: Mount the PCB terminals and connectors, followed by low-profile components (resistors and zeners), then smaller capacitors such as ceramics, and finally larger electrolytic capacitors. Ensure correct placement of zener diodes in the PCB as per schematic diagrams. In fact, you may like to fit a good-quality 8-pin DIL socket to the board to allow the amplifier IC1 to be conveniently plugged in later on.

The PCB can fit inside the box by proper drilling, reaming and cutting holes on the front and rear panels and the base of the cabinet.

Note that besides mounting holes in the rear panel for the speaker terminal, there should be clearance holes for other terminals also. On the front panel, the holes for volume control, mono/stereo switch and RCA input sockets are all circular, so they can be drilled and reamed. The holes for RCA sockets should be large enough to accept plastic insulating sleeves. Two holes have to be drilled for volume controls on the front panel.

Larger holes for the mains fuse holder (F1) and power switch (S2) are irregular. So, if you prepare them by hand, careful drilling is required besides a nibbling tool and/or small jeweller's files.

Once the wires are connected, carefully place the PCB inside the cabinet.